

浙江大学 2025–2026 学年春夏学期
《数据库系统》课程课堂测试一
(Quiz 1 for Database Systems)

Consider the following three relation schemas.

Student(*Sid*, *Name*, *Sex*, *Age*)

Participate(*ProjectName*, *Sid*, *Score*)

Project(*ProjectName*, *Requirement*, *Starttime*, *Endtime*)

1. Please write relational-algebra expressions for the following queries.

- (1) Find the names of students who are in ‘MiniSQL’ project.
- (2) Find the Sid of students who have not been in any project yet.
- (3) For each project, find the names of students who have the highest score.
- (4) Find Top-2 projects which have the most female participants.

2. Please analyze relational-algebra expression.

- (5) What’ s the semantic of the following query?

$$\Pi_{Sid} \left(\sigma_{ProjectName = (\Pi_{ProjectName} (\sigma_{Starttime = \min(Starttime)} (Project)))} (Participate) \right)$$